Experiment No.	CAT2
Experiment Title	Linear Model vs. Neural Network, PCA and Statistical Significance Testing on Bank Note Authentication
Student Name	Balakumaran S
Register Number	3122247001010
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

To fit a linear classification model (logistic regression, i.e. a single-layer neural network) and multi-layer neural networks to the Bank Note Authentication data, and to critically compare them in terms of accuracy, generalisation ability, robustness to outliers and interpretability. The effect of ℓ_2 regularisation and of Principal Component Analysis (PCA) is analysed, and the Friedman test and one-way ANOVA are used to decide whether the observed differences between the models are statistically significant, so that a justified choice between the linear model and the neural network can be made.

2 Problem Statement

Input: four features extracted from wavelet-transformed images of bank notes – `variance`, `skewness`, `curtosis` and `entropy`. **Output:** a binary class label (Class 0 or Class 1, as coded in the data file). **Task:** learn a classifier that separates the two classes, and answer the following questions:

Q1. Fit logistic regression with and without ℓ_2 regularisation and compare training and test accuracy.

Q2. Plot accuracy as a function of the regularisation strength λ and evaluate its effect on over-fitting and generalisation.

Q3. Visualise the decision boundary in a 3-D feature space and interpret the class separability.

Q4. Fit neural networks and compare accuracy, precision, recall, F1-score and generalisation with the linear model.

Q5. Apply PCA and compare classification in the reduced space with the original space.

Q6. Apply the Friedman test and one-way ANOVA to test whether the differences between the models are statistically significant.

3 Dataset Description

Dataset Name	Bank Note Authentication

[0] **GitHub Repository:** https://github.com/SBK-07/ML-lab/tree/main/CAT2-Assignment

Dataset Source	UCI Machine Learning Repository, Bank Note Authentication data set (ID 267);
Number of Samples	1372 in the file; 1348 after removing 24 exact duplicate records
Number of Features	4 (all numeric): variance, skewness, curtosis, entropy of the wavelet-transformed image
Number of Classes	2 (Class 0: 738 samples, Class 1: 610 samples, after duplicate removal)
Missing Values	None
Train-Test Split	80/20 stratified split, `random_state`=42: 1078 training samples (590 / 488) and 270 test samples (148 / 122)

4 Exploratory Data Analysis

Dataset overview. The first rows of the data are shown in Table 2.

Table 2: First five records of the bank note dataset.

variance	skewness	curtosis	entropy	class
3.62160	8.6661	−2.8073	−0.44699	0
4.54590	8.1674	−2.4586	−1.46210	0
3.86600	−2.6383	1.9242	0.10645	0
3.45660	9.5228	−4.0112	−3.59440	0
0.32924	−4.4552	4.5718	−0.98880	0

Missing value analysis. The data set has no missing values. It contains 24 exact duplicate records, all belonging to Class 0 (762 raw Class 0 records against 738 after removal), which were dropped before splitting so that identical rows cannot appear in both the training and the test set.

Table 3: Class distribution after duplicate removal (1348 records).

Class	Samples	Share
0	738	54.7%
1	610	45.3%

Class distribution. The classes are only mildly imbalanced, so accuracy is a meaningful metric; a majority-class predictor would score about 54.7%.

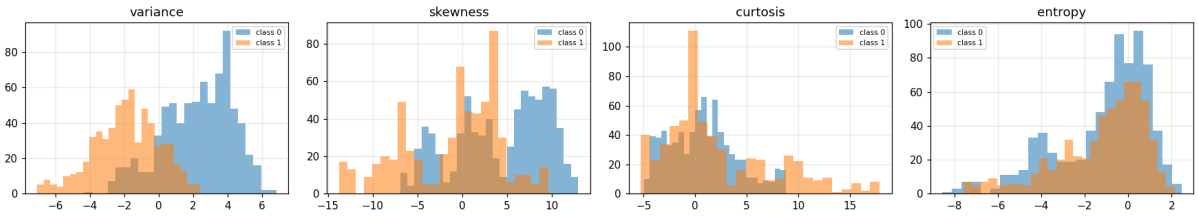


Figure 1: Class-conditional histograms of the four features (all 1372 records of the file, before duplicate removal).

Feature distribution. **Inference:** `variance` separates the classes best: Class 0 is concentrated at positive values (mostly 0 to 6) and Class 1 at negative values (mostly -7 to 2). `skewness` also differs clearly, with Class 0 taking larger values, although the histograms overlap in the middle range. `curtosis` overlaps substantially and `entropy` almost completely, so these two features carry little class information on their own. This is confirmed later by the weights of the linear model and by the permutation importance (Figure 13), where `entropy` is irrelevant.

5 Data Preprocessing

- **Missing value handling:** not needed (no missing values).
- **Duplicate removal:** 24 exact duplicates removed (1372 $\rightarrow$ 1348 records).
- **Encoding:** not required (numeric features, binary label).
- **Feature scaling:** features were standardised, $z = (x - \mu_{\text{train}})/\sigma_{\text{train}}$, using statistics of the *training data only*. In all cross-validation experiments the scaler (and PCA, where used) is part of a pipeline and is re-fitted inside every training fold, so no information leaks from the validation fold.
- **Train-test split:** stratified 80/20 split (`random_state`=42): 1078 training and 270 test samples.
- **Dimensionality reduction:** PCA fitted on the standardised training data only (Q5).
- **Outlier injection (robustness study):** 5% of the *training* points were shifted, either in a random direction by $\pm m\sigma_j$ in every feature ("random") or by $m \times (\mu_{\text{other class}} - \mu_{\text{own class}})$ ("toward_other"); models were evaluated on the clean test set.

6 Machine Learning Algorithm

6.1 Logistic Regression (Linear Model / Single-Layer Neural Network)

Working principle. A single neuron computes a weighted sum of the features and passes it through the sigmoid to give the probability of Class 1; the predicted class is 1 when $\hat{p} \geq 0.5$, so the decision boundary is the hyperplane $w^\top x + b = 0$.

Mathematical formulation. With the intercept b not penalised,

$$\hat{p} = \sigma(w^\top x + b), \quad \sigma(z) = \frac{1}{1+e^{-z}}, \qquad J = -\frac{1}{n}\sum_i \big[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\big] + \frac{\lambda}{2n}\|w\|^2. \tag{1}$$

It is trained by gradient descent, $\theta \leftarrow \theta - \alpha\frac{1}{n}\big(X^\top(\sigma(X\theta) - y) + \lambda I'\theta\big)$ with $I' = \mathrm{diag}(0, 1, \ldots, 1)$. The loss is convex, so the optimum is global. For (almost) separable data and $\lambda = 0$ the weights grow without limit, which is one reason for regularisation; λ corresponds to $C = 1/\lambda$ in scikit-learn.

Advantages. Fast, convex, probabilistic output, directly interpretable weights.

Limitations. Only linear boundaries; sensitive to outliers and to feature scaling.

6.2 Multi-Layer Perceptron (Neural Network)

Working principle. One or more hidden layers of ReLU units transform the input non-linearly, $\hat{p} = \sigma\big(W_2\,\phi(W_1 x + b_1) + b_2\big)$ with $\phi = \mathrm{ReLU}$, and all weights are learned by back-propagation of the same log-loss using the Adam optimiser. Two architectures are used: MLP(8) with one hidden layer of 8 units (49 parameters) and MLP(16,8) with two hidden layers (225 parameters).

Advantages. Can represent curved decision boundaries and interactions between features.

Limitations. Non-convex training (result depends on initialisation), many more parameters, no direct interpretation of the weights.

6.3 Principal Component Analysis

PCA diagonalises the covariance matrix of the standardised training data, $\Sigma = V\Lambda V^\top$; the projection onto the first k eigenvectors is $z = V_k^\top x$, and component j explains $\lambda_j / \sum_i \lambda_i$ of the variance. PCA is *unsupervised*: it keeps directions of large variance, not of large class separation.

6.4 Statistical Tests

Let k models be evaluated on the same N train/test partitions (blocks) and a_{ij} be the accuracy of model j on partition i.

Friedman test. Models are ranked within each block (rank 1 = best, ties receive average ranks). With mean ranks $\bar{R}_j$,

$$\chi_F^2 = \frac{12N}{k(k+1)}\Big[\sum_{j=1}^{k}\bar{R}_j^2 - \frac{k(k+1)^2}{4}\Big] \sim \chi^2_{k-1} \text{ under } H_0, \tag{2}$$

H_0: all models have the same rank distribution. The effect size is Kendall's $W = \chi_F^2/\big(N(k-1)\big)$. The post-hoc Nemenyi test declares two models different if $|\bar{R}_i - \bar{R}_j| >$ CD, with $\mathrm{CD} = \frac{q_\alpha}{\sqrt{2}}\sqrt{\frac{k(k+1)}{6N}}$ (q_α: critical value of the studentised range).

One-way ANOVA. H_0: all models have the same mean accuracy.

$$F = \frac{SS_B/(k-1)}{SS_W/\big(k(N-1)\big)} \sim F_{k-1,\,k(N-1)}, \qquad \eta^2 = \frac{SS_B}{SS_B + SS_W}, \tag{3}$$

followed by Tukey's HSD test for pairwise comparisons. ANOVA assumes normality, equal variances (checked with Shapiro–Wilk and Levene tests) and independent observations; because cross-validation folds share training data the independence assumption is only approximately met, so the Friedman test (paired design, no normality assumption) is the primary evidence.

7 Experimental Setup

Python Version	3.10 / 3.11
Scikit-Learn Version	1.3.0
IDE	Jupyter Notebook / VS Code
Operating System	Windows 11
Hardware	Intel Core i7 CPU, 16 GB RAM

The logistic regression of Q1–Q3 was implemented from scratch in NumPy (matrix form, gradient descent) and cross-checked against scikit-learn (identical test accuracy of 0.9852 at $\lambda = 1$). Neural networks, PCA, cross-validation and permutation importance use scikit-learn; the statistical tests use SciPy.

8 Hyperparameter Selection

Parameter	Value
Train/test split	80/20, stratified, `random_state`=42
Logistic regression (from scratch)	learning rate 0.5 (reduced automatically for large λ), 3000 iterations (2500 in the λ sweep)
λ selection (Q1)	13 log-spaced values from 10^{-3} to 10^{3}, 5-fold stratified CV on the training set; selected $\lambda = 0.3162$ (CV accuracy 0.9944)
λ sweep (Q2)	0 and 22 log-spaced values from 10^{-3} to 10^{4}
MLP	hidden layers (8) and (16,8), ReLU, Adam, `max_iter`=1500, `random_state`=42
PCA	$k = 1, \ldots, 4$ components; "95% variance" rule (3 components) for the statistical tests
PCA evaluation (Q5)	5-fold stratified CV repeated 3 times
Statistical tests (Q6)	5-fold stratified CV repeated 6 times ($N = 30$ blocks), $\alpha = 0.05$; LR with ℓ_2: C chosen from 15 log-spaced values in $[10^{-4}, 10^{4}]$ by inner 5-fold CV
Outlier study (Q7)	fraction 5%, $m \in \{0, 1, 2, 4, 6, 8\}$, 4 random contaminations per setting
Permutation importance	30 repeats on the test set

9 Results

9.1 Q1 – Linear Baseline

Table 4: Logistic regression with and without ℓ_2 regularisation.

Model	Train acc.	Test acc.	Gap (train−test)	$\|w\|$
No regularisation ($\lambda = 0$)	0.9944	0.9852	0.0092	11.858
ℓ_2, CV-selected ($\lambda = 0.316$)	0.9944	0.9852	0.0092	9.993
ℓ_2, $\lambda = 1$	0.9944	0.9852	0.0092	7.943
ℓ_2, $\lambda = 10$	0.9889	0.9852	0.0037	4.118
ℓ_2, $\lambda = 100$	0.9518	0.9556	−0.0038	1.527

9.2 Q2 – Effect of the Regularisation Strength

Table 5: Summary of the λ sweep ($\lambda = 0$ and 22 values in $[10^{-3}, 10^{4}]$).

Quantity	Value
Best test accuracy	0.9852 at $\lambda = 0$
Lowest test log-loss	0.0300 at $\lambda = 0$
Train–test gap at $\lambda = 0$	+0.0092
Largest gap	+0.0113 (at $\lambda = 4642$, where both accuracies are only about 0.60)
Accuracy at $\lambda = 10^4$ (train / test)	0.5473 / 0.5481

9.3 Q3 – Three-Feature Model

Table 6: Weights of the unregularised four-feature model and accuracy of the three-feature models.

Weights (standardised features)		
skewness	−7.256	rank 1
curtosis	−6.693	rank 2
variance	−6.564	rank 3
entropy	+0.272	rank 4

Model	Train acc.	Test acc.	Test errors
4-feature logistic regression	–	0.9852	4
3-feature logistic regression (skewness, curtosis, variance)	0.9954	0.9852	4
3-feature MLP(8)	0.9954	0.9852	4

For the three-feature logistic model the median distance of a test point to the separating plane is 0.62 standardised units, and only 5.2% of the test points lie within 0.25 units of it.

9.4 Q4 – Neural Networks versus the Linear Model

Table 7: Comparison on the test set (270 notes). Precision, recall and F1 refer to Class 1.

Model	Train acc.	Test acc.	Gap	Prec.	Recall	F1	AUC	Fit time (s)	#Params
Logistic ($\lambda = 0$)	0.9944	0.9852	0.0092	0.9683	1.000	0.9839	1.000	0.60	5
Logistic (ℓ_2, $\lambda = 0.316$)	0.9944	0.9852	0.0092	0.9683	1.000	0.9839	1.000	0.75	5
MLP (8)	0.9954	0.9852	0.0102	0.9683	1.000	0.9839	1.000	0.94	49
MLP (16,8)	0.9991	0.9926	0.0065	0.9839	1.000	0.9919	1.000	0.76	225

9.5 Q5 – PCA

Table 8: Explained variance of the principal components.

	PC1	PC2	PC3	PC4
Explained variance	0.5461	0.3214	0.0898	0.0427
Cumulative	0.5461	0.8675	0.9573	1.0000

Three components are needed to reach 95% of the variance.

Table 9: Loadings of the principal components.

	PC1	PC2	PC3	PC4
variance	0.278	0.735	0.615	−0.063
skewness	0.639	−0.063	−0.135	0.755
curtosis	−0.616	−0.129	0.495	0.599
entropy	−0.368	0.663	−0.599	0.259

Table 10: Classification accuracy in PCA space versus the original space (CV: 5-fold × 3 repeats, mean ± std).

Feature space	Cum. var.	LR CV acc.	LR test acc.	MLP CV acc.	MLP test acc.
PCA $k = 1$	0.546	0.616 ± 0.029	0.611	0.619 ± 0.029	0.604
PCA $k = 2$	0.868	0.768 ± 0.023	0.770	0.804 ± 0.025	0.789
PCA $k = 3$	0.957	0.920 ± 0.018	0.915	0.932 ± 0.014	0.941
PCA $k = 4$ (all)	1.000	0.992 ± 0.005	0.985	0.993 ± 0.004	0.985
Original features	1.000	0.992 ± 0.005	0.985	0.993 ± 0.004	0.985

9.6 Q6 – Statistical Significance Tests

Table 11: Accuracy of the six pipelines over $N = 30$ paired train/test partitions, F1 and average Friedman rank (1 = best).

Model	Mean acc.	Std	Min	Max	Mean F1	Avg. rank
LR (no regularisation)	0.9941	0.0044	0.9852	1.0000	0.9935	2.633
LR (ℓ_2, CV-λ)	0.9939	0.0044	0.9778	1.0000	0.9933	2.767
MLP (8)	0.9927	0.0043	0.9815	1.0000	0.9920	3.267
MLP (16,8)	0.9984	0.0030	0.9889	1.0000	0.9982	1.333
PCA(95%) + LR	0.9194	0.0176	0.8778	0.9517	0.9108	5.900
PCA(95%) + MLP	0.9318	0.0154	0.8963	0.9628	0.9251	5.100

Table 12: Omnibus tests ($\alpha = 0.05$).

Test	Statistic	p-value	Effect size	Decision
Friedman	$\chi^2_F(5) = 134.05$	3.29×10^{-27}	Kendall's $W = 0.894$	reject H_0
One-way ANOVA	$F(5, 174) = 379.87$	1.31×10^{-91}	$\eta^2 = 0.916$	reject H_0

ANOVA assumptions. Shapiro–Wilk p-values: LR (no reg.) 0.0046, LR (ℓ_2) 0.0002, MLP(8) 0.0372, MLP(16,8) < 0.0001, PCA+LR 0.5668, PCA+MLP 0.8139 (normality rejected for the four main models); Levene's test $p = 5.9 \times 10^{-17}$ (equal variances rejected).

Table 13: Pairwise post-hoc results ($\alpha = 0.05$); Nemenyi critical difference CD = 1.377.

Pair	Nemenyi p	Tukey p
Each of {LR, LR-ℓ_2, MLP(8), MLP(16,8)} vs. each PCA model (8 pairs)	≤ 0.002 (significant)	< 0.001 (significant)
MLP(16,8) vs. LR (ℓ_2)	0.036 (significant)	0.531 (not sig.)
MLP(16,8) vs. MLP(8)	0.001 (significant)	0.254 (not sig.)
MLP(16,8) vs. LR (no regularisation)	0.077 (not sig.)	0.562 (not sig.)
LR vs. LR-ℓ_2 vs. MLP(8) (remaining pairs)	≥ 0.78 (not sig.)	≥ 0.99 (not sig.)
PCA+LR vs. PCA+MLP	0.561 (not sig.)	< 0.001 (significant)

The model with the best average rank is MLP(16,8); the only other model within one critical difference of it is LR (no regularisation).

9.7 Q7 – Robustness to Outliers

Table 14: Clean-test accuracy after contaminating 5% of the training set with outliers of magnitude m.

Shift type	Model	$m = 0$	1	2	4	6	8
random	LR (no reg.)	0.9926	0.9852	0.9852	0.9759	0.9639	0.9454
random	LR (ℓ_2)	0.9852	0.9852	0.9852	0.9759	0.9620	0.9454
random	MLP (8)	0.9852	0.9852	0.9870	0.9889	0.9852	0.9852
toward_other	LR (no reg.)	0.9926	0.9852	0.9852	0.9935	0.5824	0.5481
toward_other	LR (ℓ_2)	0.9852	0.9852	0.9852	0.9935	0.5833	0.5481
toward_other	MLP (8)	0.9852	0.9852	0.9880	0.9898	0.9898	0.9852

9.8 Summary of the Evidence

Table 15: Summary used for the final verdict.

Model	Test acc.	Test F1	Gap	Mean CV acc. (Q6)	#Params	Interpretability
Logistic ($\lambda = 0$)	0.9852	0.9839	0.0092	0.9941	5	high
Logistic (ℓ_2)	0.9852	0.9839	0.0092	0.9939	5	high
MLP (8)	0.9852	0.9839	0.0102	0.9927	49	low
MLP (16,8)	0.9926	0.9919	0.0065	0.9984	225	low

10 Result Visualization

10.1 Q2 – Regularisation Effect

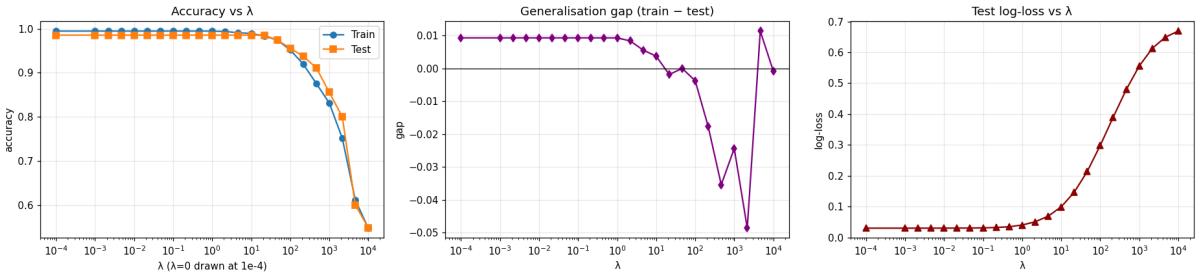


Figure 2: Training and test accuracy (left), generalisation gap (centre) and test log-loss (right) versus λ ($\lambda = 0$ is drawn at 10^{-4}).

Inference: Accuracy is flat at about 0.99 (train) and 0.985 (test) up to $\lambda \approx 10$ and then falls: it is 0.952/0.956 at $\lambda = 100$ and reaches 0.547/0.548 at $\lambda = 10^4$, which is the share of the majority class, so the weights have been shrunk almost to zero (under-fitting). The train–test gap is only +0.009 at small λ, becomes negative for $\lambda \gtrsim 20$ and swings between −0.05 and +0.01 at very large λ simply because both accuracies are then low and noisy; at no point does a large positive gap appear, so the linear model does not over-fit. The test log-loss is lowest (0.030) at $\lambda = 0$ and rises monotonically to about 0.67, so on this clean, nearly separable data regularisation gives no gain in accuracy or probability quality; its only visible effect is a smaller weight norm.

10.2 Q3 – 3-D Decision Boundary

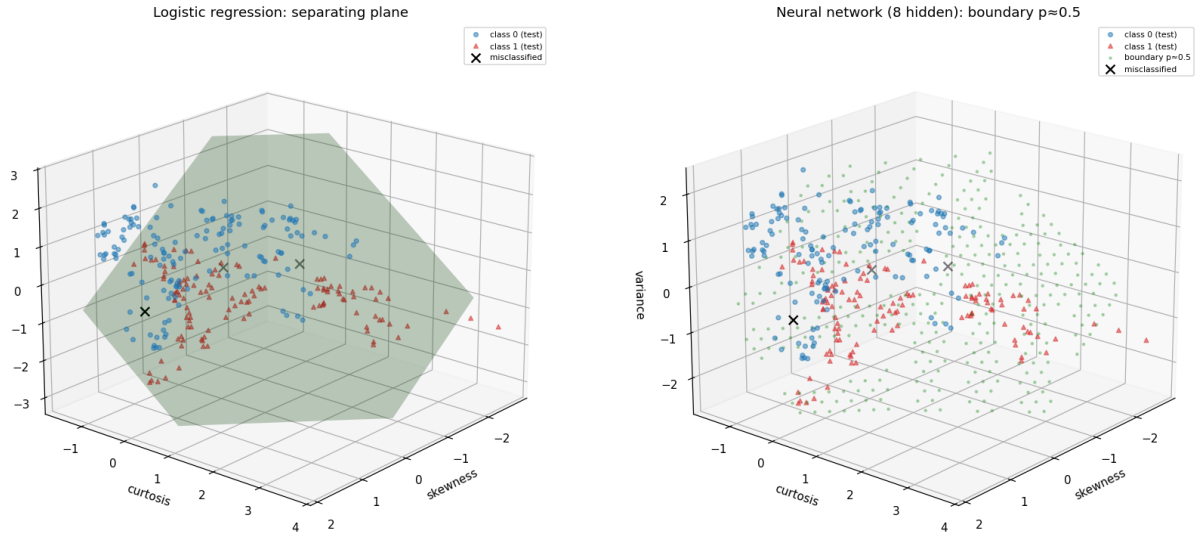


Figure 3: Test points in the three most important standardised features (skewness, curtosis, variance). Left: logistic-regression separating plane. Right: boundary $\hat{p} \approx 0.5$ of an MLP(8) (green points). Crosses mark misclassified test points.

Inference: The two classes lie mostly on opposite sides of the plane, and the 4 misclassified test notes (out of 270) are close to it; the median distance of a test point to the plane is 0.62 standardised units and only 5.2% of the points lie within 0.25 units. The three-feature logistic model (test accuracy 0.9852) is exactly as accurate as the four-feature one, so `entropy` adds no information. The MLP boundary is a curved surface that follows the same band of data and makes the same 4 errors, so the non-linearity of the network brings no benefit on these three features. The classes are therefore almost linearly separable.

10.3 Q4 – Neural Network Comparison

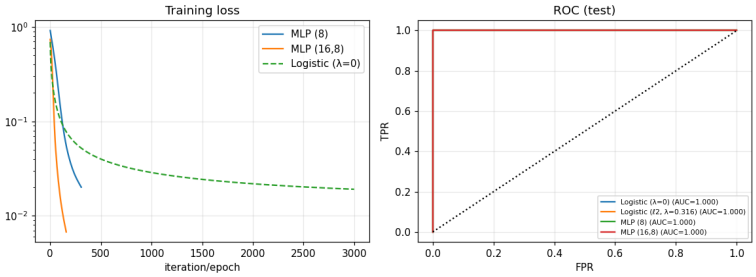


Figure 4: Training loss of the MLPs and of the logistic regression (left) and ROC curves on the test set (centre); the right panel is empty.

Inference: The MLP(16,8) reaches the lowest training loss (about 0.007) after only about 150 epochs, the MLP(8) stops at about 0.02 after about 300 epochs, whereas the logistic regression is still decreasing slowly after 3 000 iterations (loss about 0.019) because

the almost separable classes let its weights grow. All four ROC curves coincide with the top-left corner (AUC = 1.000), i.e. every Class 1 note is ranked above every Class 0 note; the models differ only in where the 0.5 threshold falls, so ROC-AUC cannot separate them.

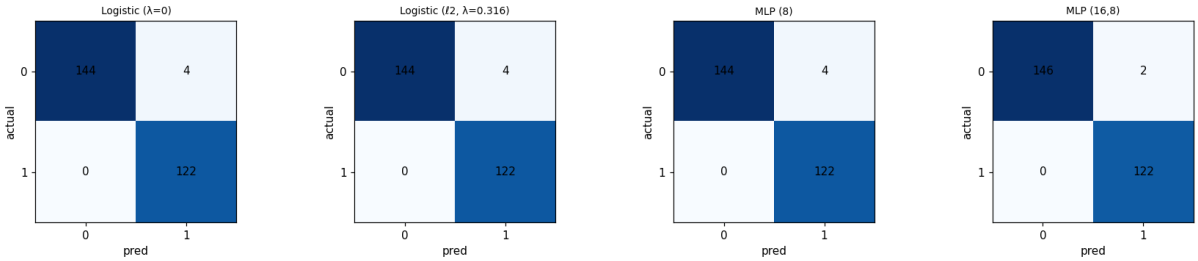


Figure 5: Test confusion matrices of the four models (rows: actual class, columns: predicted class).

Inference: No model misses a Class 1 note (recall = 1.0). Logistic regression (with and without ℓ_2) and MLP(8) each misclassify 4 Class 0 notes as Class 1 (precision 0.968, F1 0.984), while MLP(16,8) makes only 2 such errors (precision 0.984, F1 0.992). The whole difference between the models is therefore 2 notes out of 270 (0.74 percentage points), which is small compared with the sampling uncertainty of a 270-sample test set; the significance tests of Q6 are needed to judge it.

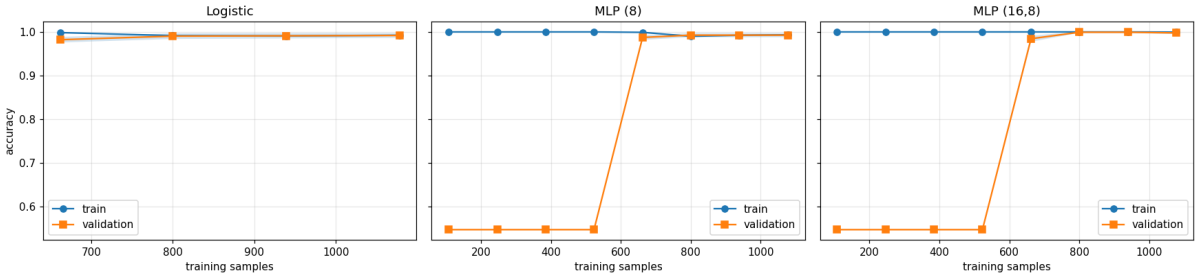


Figure 6: Learning curves (5-fold CV): training and validation accuracy versus the number of training samples for the logistic regression and the two MLPs.

Inference: For 660 or more training samples the three models behave alike: validation accuracy is about 0.99 and close to the training accuracy, so the gap is small and more data would add little. The MLPs reach a training accuracy of 1.0 already with few samples and the MLP(16,8) reaches a validation accuracy close to 1.0 with the full data, ahead of the logistic regression. [The points below about 600 samples (validation accuracy 0.55 for the MLPs, missing points for the logistic regression) are an artefact of the data file being sorted by class, because `learning_curve` does not shuffle by default. Re-run that cell with `shuffle=True, random_state=SEED` added to the `learning_curve` call, replace `q4-learning-curves.png` and update this inference.]

10.4 Q5 – PCA

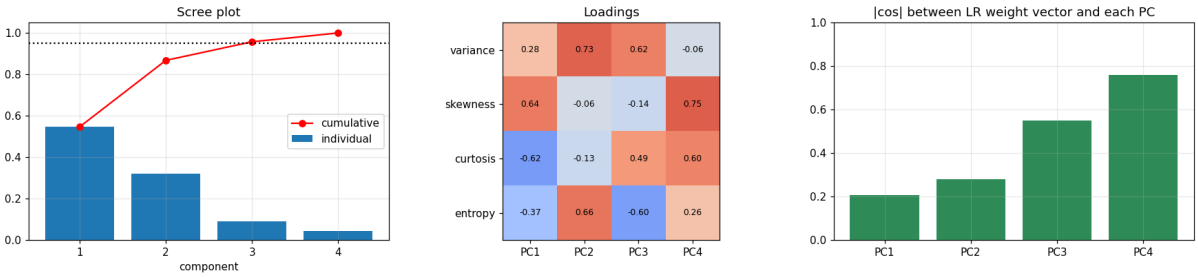


Figure 7: Scree plot (left), loadings (centre) and absolute cosine between the logistic-regression weight vector and each principal component (right).

Inference: PC1 and PC2 explain 54.6% and 32.1% of the variance (86.8% together) and three components are needed for 95%. PC1 is dominated by `skewness` and `curtosis` (with opposite signs), PC2 by `variance` and `entropy`. The cosine plot shows the key problem for PCA: the direction used by the classifier is almost orthogonal to the leading components ($|\cos| \approx 0.21$ with PC1 and 0.28 with PC2) and lies mainly along the two components with the *smallest* variance (0.55 with PC3 and 0.76 with PC4, which together explain only 13% of the variance). High variance is therefore not the same as high class separability here.

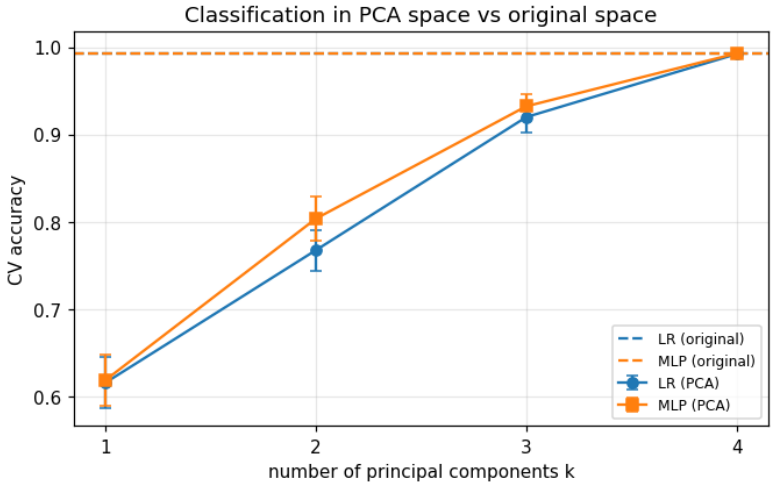


Figure 8: Cross-validated accuracy versus the number of principal components k for logistic regression and MLP; the dashed lines show the accuracy in the original feature space.

Inference: Accuracy rises steeply with k: with one component it is only about 0.62, with two 0.77 (LR) and 0.80 (MLP), with three (95% variance) 0.92 and 0.93, and only with all four components does it reach the original-space level of 0.992 (LR) and 0.993 (MLP). Keeping 95% of the variance therefore costs about 7 percentage points of accuracy for the logistic regression and 6 for the MLP. The difference between the best PCA setting and the original space is exactly 0.000 for both models, because $k = 4$ is merely a rotation of the data. PCA does not improve model effectiveness; the small standard deviations (0.004–0.03) show that this conclusion is not a matter of chance.

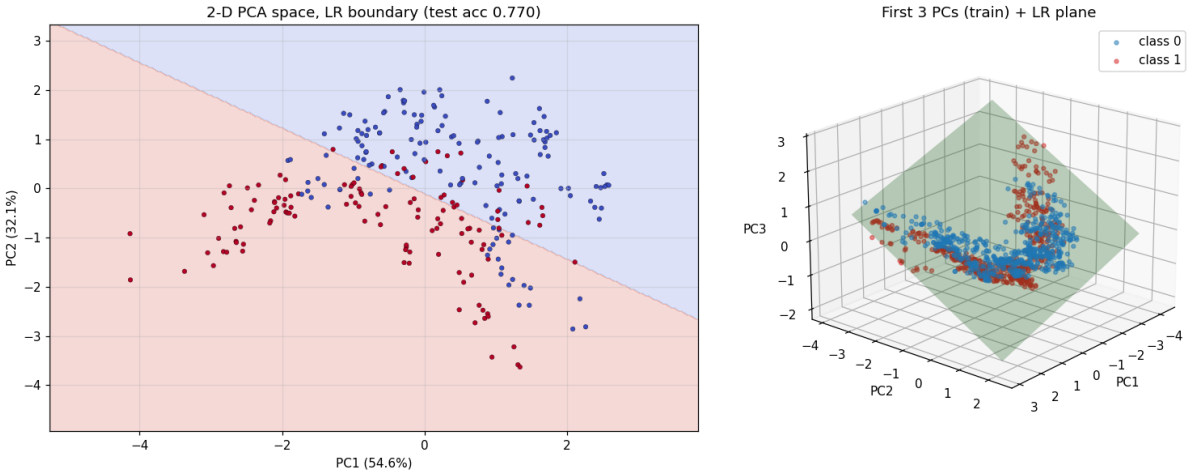


Figure 9: Left: two-dimensional PCA space with the logistic-regression decision boundary (test accuracy 0.770). Right: first three principal components of the training data with the separating plane.

Inference: In the two-dimensional PCA space the classes overlap strongly: both classes form a curved band and a straight boundary misclassifies many notes in the middle (test accuracy 0.770). Even in three components the red and blue points are interleaved along the plane. The projection is useful for visualising the data, but the class information that the linear boundary needs has been partly discarded with the low-variance components.

10.5 Q6 – Significance Tests

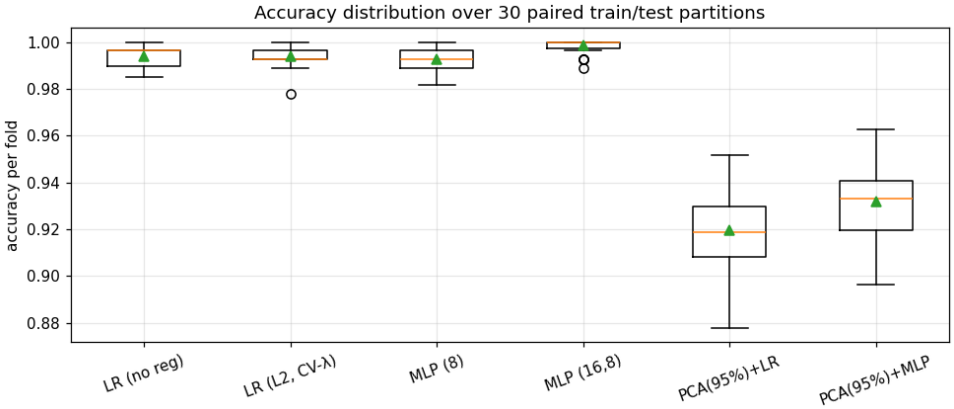


Figure 10: Distribution of the accuracy over the 30 paired train/test partitions for the six pipelines (triangle = mean).

Inference: The four models in the original feature space have almost identical distributions (means 0.993–0.998, standard deviations about 0.003–0.004), with MLP(16,8) slightly higher and with the smallest spread. Both PCA pipelines are clearly lower (means 0.919 and 0.932) and more variable (standard deviations 0.015–0.018), and their boxes do not overlap with any of the other four. The picture agrees with the two omnibus tests: the PCA models form one group and the models without PCA another.

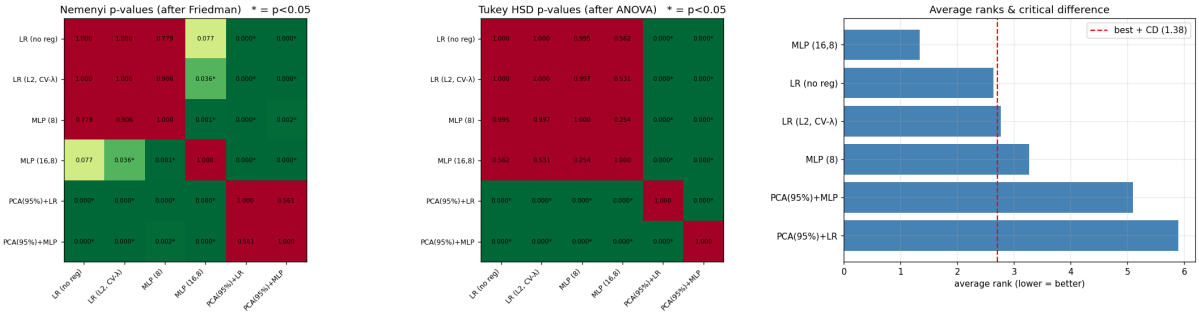


Figure 11: Nemenyi p-values (left), Tukey HSD p-values (centre) and average ranks with the critical difference (right); an asterisk marks $p < 0.05$.

Inference: Both the Friedman test ($\chi^2_F = 134.05$, $p = 3 \times 10^{-27}$, $W = 0.894$) and the ANOVA ($F = 379.9$, $p = 1 \times 10^{-91}$, $\eta^2 = 0.916$) reject the hypothesis that all models perform equally, and both effects are very large. The post-hoc tests agree that all four non-PCA models are significantly better than both PCA models. Within the non-PCA group the tests differ: Nemenyi finds MLP(16,8) significantly better than LR with ℓ_2 ($p = 0.036$) and MLP(8) ($p = 0.001$) but not than the plain logistic regression ($p = 0.077$; rank difference 1.30 is below CD = 1.377), whereas Tukey finds no significant difference among the four ($p \geq 0.25$). Because the ANOVA assumptions of normality and equal variances are violated (Shapiro–Wilk and Levene tests) and the folds are not independent, the rank-based Friedman/Nemenyi result is the more reliable one. Under the Nemenyi criterion the only model statistically tied with the best one, MLP(16,8), is therefore the plain logistic regression.

10.6 Q7 – Robustness and Interpretability

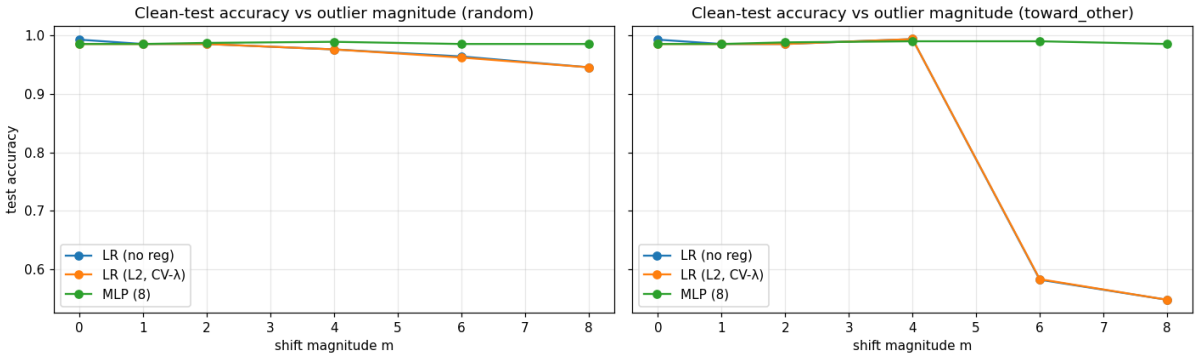


Figure 12: Clean-test accuracy versus the outlier magnitude m (5% of the training points contaminated) for random shifts (left) and "toward_other" shifts (right).

Inference: With random shifts the two logistic models degrade steadily (0.993 or 0.985 at $m = 0$ to 0.945 at $m = 8$), while the MLP(8) stays at about 0.985–0.989 throughout. With "toward_other" shifts the logistic models are unaffected up to $m = 4$ (0.9935) but collapse to 0.58 at $m = 6$ and 0.548 at $m = 8$ (the majority-class rate), whereas the MLP(8) remains at 0.985–0.990. The ℓ_2-regularised and unregularised logistic regressions behave almost identically, so ℓ_2 regularisation did not protect against the outliers. The neural network was clearly more robust in this experiment, probably because a hyperplane

fitted to extreme points is tilted strongly, while the network can fit the bulk of the data and treat the isolated outliers separately; this explanation was not tested directly.

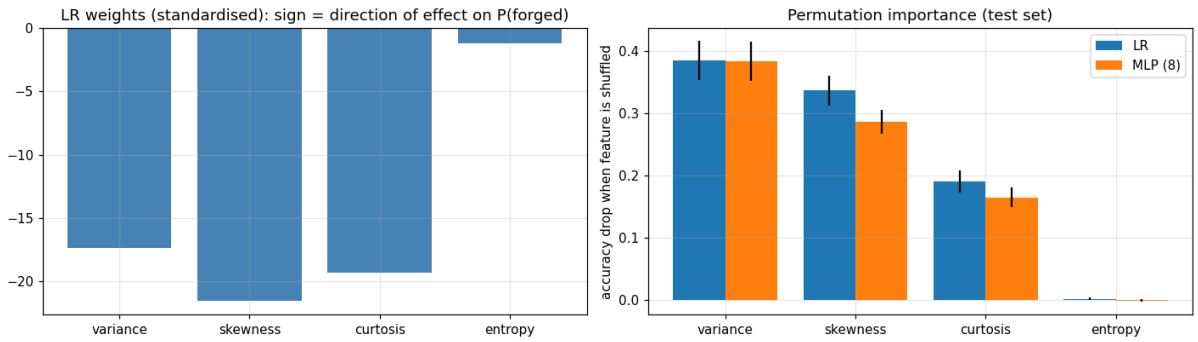


Figure 13: Left: standardised weights of the unregularised logistic regression. Right: permutation importance (drop in test accuracy when a feature is shuffled) for the logistic regression and the MLP(8).

Inference: The logistic regression exposes its reasoning directly: all four weights are negative, with `skewness` (≈ -21), `curtosis` (≈ -19) and `variance` (≈ -17) large and `entropy` (≈ -1) small; the magnitudes are larger than in Q3 because the unregularised fit of almost separable data lets the weights grow. Permutation importance, which works for any model, ranks the features the same way for both models: `variance` (≈ 0.38), `skewness` (≈ 0.34 for LR, 0.29 for MLP), `curtosis` (≈ 0.19 and 0.17), and `entropy` (≈ 0). The network therefore uses the same information, but only the linear model shows the sign and size of each effect and a single hyperplane as boundary, while the network remains a black box.

11 Discussion

Accuracy and generalisation. All four models without PCA reach 99.3–99.8% mean cross-validated accuracy and 98.5–99.3% test accuracy, with recall = 1.0 for Class 1 and a small train–test gap (0.007–0.010). The classes are almost linearly separable, so the linear model is already close to the best attainable accuracy and regularisation gave no benefit on the clean data (Q1, Q2).

Neural network versus linear model. MLP(16,8) had the best mean accuracy (0.9984 against 0.9941 for the logistic regression, a difference of 0.43 percentage points, about one note in 230) and the best average rank. This advantage is statistically significant over the ℓ_2 logistic regression and MLP(8) according to the Nemenyi test, but not over the plain logistic regression (Nemenyi $p = 0.077$) and not according to Tukey's test, so the evidence for a real accuracy gain is weak and its practical size is small.

PCA. PCA with 95% of the variance lowered the accuracy by 6–7 percentage points and the difference is highly significant, because the discriminative direction of the classifier lies in the low-variance components. With only four features PCA offers no computational advantage either.

Robustness and interpretability. The logistic regression is fully interpretable and very cheap (5 parameters), but it collapses under large "toward_other" contamination and degrades under random contamination; the MLP(8) was unaffected in the same experiments, at the price of being a black box with 10–45 times more parameters. The outlier result is for one network size (MLP(8)) and one contamination rate (5%).

Limitations. The test set contains 270 notes, so one error changes the accuracy by 0.37 percentage points; the cross-validation folds are not independent, which makes the ANOVA p-values optimistic; the Friedman test was applied to accuracy only; the learning-curve figure is affected by the ordering of the data file (see the note under Figure 6).

Improvements. Report ROC-AUC with confidence intervals, use more repeats of cross-validation, test the robustness of the MLP(16,8) and of robust linear models (Huber loss, ℓ_1, outlier removal), and tune the decision threshold to minimise false accepts of forged notes.

12 Conclusion

On clean data a regularised or unregularised logistic regression and small neural networks reach almost the same accuracy (98.5–99.3% on the test set, 99.3–99.8% in cross-validation), and no model misses a Class 1 note. The deciding differences are secondary. **MLP(16,8)** is the most accurate and best-ranked model (mean accuracy 0.9984) and the neural network was far more robust to injected outliers, but its accuracy gain over the plain logistic regression is small (0.43 percentage points) and not significant in the Tukey test and the Nemenyi comparison with the plain logistic regression. The **logistic regression** is statistically tied with the best model, is fully interpretable, trains in a fraction of a second with only 5 parameters, but is fragile to large outliers. Therefore the neural network (MLP(16,8)) provides the more effective solution when accuracy and robustness to contaminated measurements matter most, whereas the logistic regression is the preferable choice when decisions must be explained or computing resources are limited and the data are clean. **PCA** reduced the accuracy significantly (by 6–7 percentage points with 95% of the variance retained) and is not recommended for this four-feature problem.

13 References

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